

$$\textcircled{1} \quad \underline{\underline{6.5}} - \underline{\underline{25}}$$

$$A = L D L^T \quad \rightarrow \quad (L \sqrt{D}) \sqrt{D} L^T$$

$$\rightarrow C^T C$$

$$\underline{\underline{C^T = L \sqrt{D}}}$$

$$\# \quad C = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \quad C^T = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

$$A = C^T C = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$= \underline{\underline{\begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}}}$$

$$\# \quad A = \begin{bmatrix} 4 & 8 \\ 8 & 25 \end{bmatrix} \quad C = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$C^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$\Rightarrow A = L D U$$

$$A = C^T C$$

$$C^T = L \sqrt{D}$$

$$A = \begin{bmatrix} 4 & 8 \\ 8 & 28 \end{bmatrix}$$

$$\downarrow \begin{matrix} L^{-1} & A & U \end{matrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 8 \\ 8 & 28 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 0 & 9 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 0 & 3 \end{bmatrix}$$

② 6.5-26

$$A = C^T C$$

$$C^T = L \sqrt{D}$$

i) $A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 8 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} U$$

$$C = \sqrt{D} L^T$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\text{ii)} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{bmatrix}$$

$$0 \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & \sqrt{5} \end{bmatrix}$$

③ 6.5-27

$$A = L D L^T$$

$$x^T A x = x^T L D L^T x$$

$\Rightarrow$

$$\begin{bmatrix} x + \frac{b}{a}y & y \end{bmatrix}$$

$$\begin{bmatrix} x + \frac{b}{a}y \\ y \end{bmatrix}$$

$$\boxed{\frac{(a c - b^2)}{a}}$$

④ 6.5-29

$$i) \quad \frac{\partial^2 F}{\partial x^2} = 3x^2 + 2y$$

$$\frac{\partial^2 F}{\partial x \partial y} = 2x$$

$$\frac{\partial^2 F}{\partial y^2} = 2$$

$$\frac{\partial F}{\partial x} = x^3 + 2xy = 0$$
$$x(x^2 + 2y) = 0$$

$$\frac{\partial F}{\partial y} = 2y + x^2 = 0$$

$$x = 0$$
$$x^2 = -2y$$

$y = -\frac{x^2}{2}$

⑧ 6-5-32

$$a) \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 4 & 1 \end{bmatrix}$$

Not a group.

$$b) \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Q^{-1} = Q^T \rightarrow \text{orthogonal as well}$$

$$A^{-1} = \text{orthogonal}$$

$$A = (C^T)^T$$

$$B = \begin{bmatrix} 1 & 1 & \dots \\ b_1 & b_2 & \dots \end{bmatrix}$$

$$C^T = \begin{bmatrix} 1 & 1 & \dots & 1 \\ q_1 & q_2 & \dots & q_n \end{bmatrix}$$

$$\boxed{q_i \cdot q_i = 0}$$

$$\begin{bmatrix} q_1 \cdot b_1 & q_2 \cdot b_1 & \dots \\ q_1 \cdot b_2 & q_2 \cdot b_2 & \dots \\ q_1 \cdot b_n & q_2 \cdot b_n & \dots \end{bmatrix}$$

$$\boxed{AB} \quad \text{Yes}$$

c) e^{At} exponentials.

$$e^{At} \cdot e^{Bt} = e^{(A+B)t} \quad \checkmark$$

$$(e^{At})^{-1} = e^{(-A)t} \quad \checkmark$$

Yes.

d) $P \rightarrow$ +ve eigenvalues.

$$A = S \Lambda S^{-1}$$

$$B = S_0 \Lambda_0 S_0^{-1}$$

nope

e) $D \quad \det D = 1$

$$\det(A B) = \det A \cdot \det B = 1$$

$$\det(A^{-1}) = \frac{1}{\det A} = 1$$

yes

6 6.5-33

$$Ax = \lambda x$$

$$Bx = \lambda x$$

$$ABx = \lambda x$$

$$(ABx)^T Bx = \lambda x^T Bx$$

$$x^T \boxed{(AB)^T B} x$$

=

$$\lambda \boxed{x^T Bx}$$

we always

since

B is positive
definite

$$(Bx)^T \underbrace{A^T}_{=A} (Bx)$$

also true

has to be true

$$\textcircled{7} \quad \underline{6.5 - 34}$$

$$-1, 2, -1, \quad n \times n \quad K$$

$$\frac{d^2 y}{dx^2} = \lambda y$$

$$y_0 = 0$$

$$y_1 = 0$$

$$y = \sin cx$$

$$\frac{d^2 y}{dx^2} \Rightarrow c \cos cx \Rightarrow -c^2 \sin cx$$

$$c = \pi$$

$$y = \sin k\pi x$$

$$\boxed{-k^2 \pi^2}$$

⑧ 6.5-35

$$y^T C y > 0$$

$Ax \neq 0$ whenever $x \neq 0$
↓
non-singular.

$$(Ax)^T C (Ax)$$

$$x^T \underline{(A^T C A)} x$$

⑨ 8.1-3

$$A^T C A u = 0$$

$$\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$A^T C A u = f = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$A^T C A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} = B$$

$$B u = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$N(A) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$F_3 = 1$$

$$u = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{for } \begin{matrix} c_2 = 1 \\ c_3 = 1 \end{matrix}$$

10) 8-1-8

$A \rightarrow n \times n$ non-singular rank n

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^T y = \frac{1}{\Delta x} A^T y \approx -\frac{dy}{dx}$$

$$\Rightarrow \frac{1}{\Delta x} A^T y \quad y = -\int_1^x f(x) dx$$

$$y = (-x + c)$$

$$c = 1$$

$$y = 1 - x$$

